A Generalized Method for Solving a Class of Ring Coloring Problems

Tony,Guo

September 6, 2025

1 Introduction

There is a particular class of coloring problems that can be transformed, without altering the intrinsic properties of the original figure, into a special ring form. Such problems are often solved by case-by-case analysis; however, as the number of colored regions increases, this approach becomes troublesome and inefficient. Therefore, I propose a general method that applies to all coloring problems of this type.

2 Before the Ring: Linear Form

For the linear form listed in Figure 1 below, if we have $k(k \geq 2)$ colors to color n areas on a line and require the adjacent areas cannot have the same color. If we do not have to use all k colors, then apparently the total number of coloring strategies will be $U = k \cdot (k-1)^{n-1}$.

Especially if each color has to be used at least once, then we have to apply the principle of inclusion-exclusion.

$$\left| \bigcup_{j=1}^{n} A_j \right| = \sum_{h=1}^{n} (-1)^{h+1} \left(\sum_{1 \leq j_1 < \ldots < j_h \leq n} |A_{j_1} \cap \ldots \cap A_{j_k}| \right) \tag{1}$$

Define the event $A_j := \{\text{the coloring sequence of length } n \text{ does not contain color}\{c_i\}\}$ Now, suppose that exactly i colors are excluded from use. The remaining $k-i$ colors must then be used to color the n regions, which gives $(k-i) \cdot (k-i-1)^{n-1}$ possible sequences. Since there are $\binom{k}{i}$ ways to choose the excluded colors, the principle of inclusion-exclusion yields the following total number $N(n,k)$ of admissible coloring that employ all k colors at least once:

$$\tag{2}$$

$$\sum_{1 \leq i_1 < \ldots < i_h \leq n} |A_{i_1} \cap \ldots \cap A_{i_k}| = \binom{k}{i} (k-i) \cdot (k-i-1)^{n-1} \tag{3}$$

$$N(n,k) = |U| - \left| \bigcup_{i=1}^{k} A_i \right| \tag{4}$$

$$= \sum_{i=0}^{k} (-1)^i \binom{k}{i} (k-i)(k-i-1)^{n-1} \tag{5}$$

Figure 1: The areas in a line form

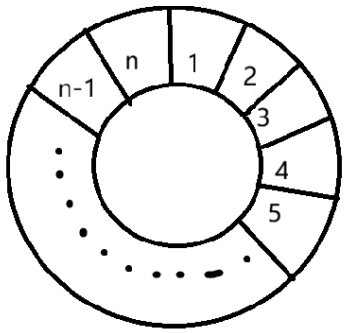

Figure 2: A general ring form pattern

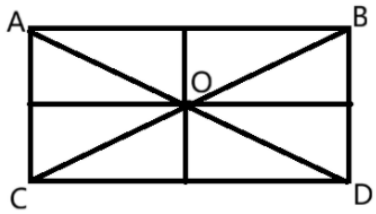

Figure 3: The O in the middle can be broken

3 Ring Form

3.1 The Shape Can Be Transformed Into the Ring

All figures that can be transformed, through certain scaling or stretching operations, into the ring-shaped structure illustrated in Figure 2 can be classified as circular coloring diagrams. Such transformations may also involve the opening of certain nodes that do not introduce adjacency between previously non-adjacent regions, as in the case of point O in Figure 3. Several examples of diagrams satisfying this property are shown in Figure 4.

Furthermore, If the regions with non-zero area in such diagrams are simplified into line segments, with their connections represented by nodes placed at the endpoints, the ring-shaped diagrams can be further reduced to node-line graphs, as illustrated in Figure 5. This offers a more concise form of simplification, which may allow the application of broader analytical tools to derive additional properties of these coloring diagrams. A detailed exploration of this direction is beyond the scope of my work, that I won't talk much about that in this passage.

3.2 The Ring Containing an Non-colored Central Hollow

We now proceed to derive the general formula in the simplest setting. As illustrated in Figure 6, suppose the circular diagram consists of n regions, where region 1 is adjacent to region n. The central hole of the ring is left uncolored. A total of k colors are available for using, and it is not required that all colors must be used. Let P_n denote the number of admissible colorings under these conditions.

We begin by examining region 1 and considering the coloring of region $(n-1)$, which is separated from region 1 by a single intermediate region n.

Case 1. If regions 1 and $(n-1)$ are assigned the same color, then the intermediate region n can be colored in $(k-1)$ ways. In this situation, regions $(n-2)$ and $(n-1)$ must be colored differently, which implies that regions 1 $(n-2)$ cannot share the same color once the diagram is reduced to

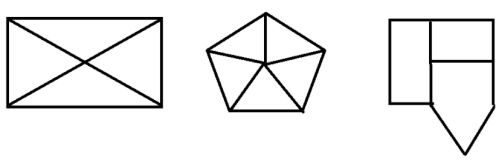

Figure 4: Some diagrams can be tranformed into a ring.

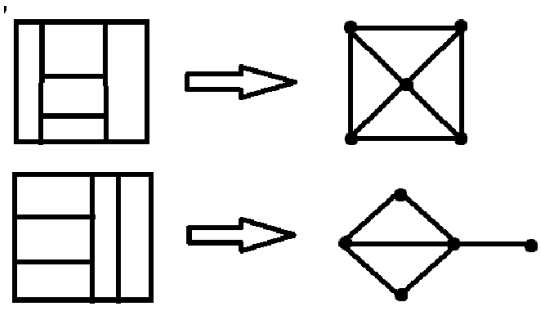

Figure 5: Transform into the node-line pattern

a smaller circular configuration. This reduction yields a ring of size $(n-2)$ and hence there are P_{n-2} admissible coloring strategies. The total contribution of this case is therefore $(k-1)P_{n-2}$.

Case 2. If regions 1 and $(n-1)$ are assigned different colors, then by a similar argument, the total number of admissible coloring strategies is $(k-2)P_{n-1}$.

By the addition principle, we thus obtain the second-order linear homogeneous recurrence relation linking P_n,P_{n-1}, and P_{n-2}

$$P_n = (k-1)P_{n-2} + (k-2)P_{n-1} \tag{6}$$

Next, we aim to derive the closed-form expression for this sequence. There are multiple ways to solve for the closed-form expression, including the method of characteristic equations and roots, which often regarded as a special case of the method of difference equations. I will use the generating function to solve for $P(n)$

Prior to solving, it is necessary to make explicit the initial conditions of the sequence. Apparently, $(n-2) > 0$, so $n \geq 3$. Our recurrence relation only works for $n \geq 3$. For the first two terms of this sequence, P_3 and P_4, we can draw the pattern and easily figure out that $P_3 = k(k-1)(k-2)$ and $P_4 = k(k-1)\left[(k-1) + (k-2)^2\right]$. Notice that P_4 needs to be discuss in cases, similar approach as we used in finding the recurrence relation above.

$$P_3 = k(k-1)(k-2); P_4 = k(k-1)\left[(k-1) + (k-2)^2\right] \tag{7}$$

The generating function $G(x)$ of P_n is a formal power series.

$$G(x) := \sum_{n=3}^{\infty} P_n x^n \tag{8}$$

Next, we perform summation on both sides of the recurrence relation, with a slight adjustment of the summation indices to achieve a more concise form. The resulting summations are then expressed in terms of $G(x)$. By substituting the previously obtained closed-form expressions for P_3 and P_4 and subsequently simplifying, we derive an explicit algebraic representation of $G(x)$. Then factor out x^3 and define a new intermediate function $H(x)$ to make the expression clear.

$$\sum_{n=5}^{\infty} P_n x^n = (k-1)\sum_{n=5}^{\infty} P_{n-2}x^n + (k-2)\sum_{n=5}^{\infty} P_{n-1}x^n \tag{9}$$

$$G(x) - P_3x^3 - P_4x^4 = (k-1)x^2G(x) + (k-2)x[G(x) - P_3x^3] \tag{10}$$

$$G(x)[1-(k-1)x^2-(k-2)x] = x^4[P_4-(k-2)P_3] + P_3x^3 \tag{11}$$

$$G(x) = x^3\left[\frac{x(k(k-1)^2)+P_3}{1-(k-2)x-(k-1)x^2}\right] \tag{12}$$

$$H(x) := \left[\frac{x(k(k-1)^2)+P_3}{1-(k-2)x-(k-1)x^2}\right] \tag{13}$$

$$G(x) = x^3H(x) \tag{14}$$

Next, the method of undetermined coefficients is employed to decompose $H(x)$ into the sum of two fractional terms, where A and B are constants (depending on k) within the function $H(x)$. By equating coefficients after bringing the fractions to a common denominator, we compare the constant term and the coefficient of x in the numerator, which yields a system of two linear equations in A and B.

$$H(x) = \frac{A}{1-(k-1)x} + \frac{B}{1+x} \tag{15}$$

$$\begin{cases} A + B = k(k-1)(k-2) \\ A - B(k-1) = k(k-1)^2 \end{cases} \tag{16}$$

By observing and performing elimination on this system, it follows that $A = (k-1)^3$, $B = -(k-1)$. Note that we had the order restriction of $n-3 \geq 0$, so we substitute $;m = n-3$ to make things clear. By applying the power series,we have:

$$H(x) = \frac{(k-1)^3}{1-(k-1)x} - \frac{k-1}{1+x} \tag{17}$$

$$= (k-1)^3 \sum_{m\geq 0} (k-1)^m x^m - (k-1) \sum_{m\geq 0} (-1)^m x^m \tag{18}$$

$$G(x) = \sum_{m\geq 0} (k-1)^{m+3} x^{m+3} - \sum_{m\geq 0} (k-1)(-1)^m x^{m+3} \tag{19}$$

Take the coefficient of the generating function, we derive the closed-form expression:

$$P_n = (k-1)^n - (k-1)(-1)^{n-3} \tag{20}$$

$$= (k-1)^n + (k-1)(-1)^n \tag{21}$$

3.3 The Ring Including the Central Region to Be Colored

In this case, we still have k colors (Also, there is no restriction of using all the colors at least once), but the central area of the ring has to be colored as well. We now have n regions in total and assign the central one is the nth region. $(n-1)$ regions now make up a closed ring. If the central area of the ring is no longer a hollow, then we can pick one color from all k colors to paint the central area, and use the rest $(k-1)$ colors to paint the rest $(n-1)$ areas, which goes back the the former situation. After that we just have to follow the Product Rule to do the multiplication and derive the final result.

$$P(n,k) = \binom{k}{1} \left[(k-2)^{n-1} + (k-2)(-1)^{n-1}\right] \tag{22}$$

3.4 The Situation of Each Color Must Be Used at least Once

If we add the restriction that each color must be used at least once, then by using the similar approach as *Section* 2, we can derive that:

$$\sum_{1\leq i_1<\ldots<i_h\leq n} |A_{i_1} \cap \ldots \cap A_{i_k}| = \binom{k}{i} \left[(k-i-1)^n + (-1)^n (k-i-1)\right] \tag{23}$$

$$P(n,k) = |U| - \left|\bigcup_{i=1}^{k} A_i\right| \tag{24}$$

$$= \sum_{i=0}^{k} (-1)^i \binom{k}{i} \left[(k-i-1)^n + (-1)^n (k-i-1)\right] \tag{25}$$

4 Remark

4.1 The Connection Between the Ring Form and the Linear Form

The formula (21) can be derived in another easier method. The total number of colors is still k and we don't have to use all the colors. We can cut the ring at any point where two distinct regions meet, unfolding it into a linear arrangement. When the colors at the two ends of this linear arrangement are identical, joining the ends will result in two adjacent regions having the same color, while all other adjacent regions have distinct colors. If the two adjacent regions with the same color are then merged into a single region, the resulting ring has $n-1$ regions, with all adjacent regions colored differently, corresponding to P_{n-1} coloring configurations. When the colors at the two ends are different, joining the ends produces a ring with n regions, all adjacent regions having distinct colors, corresponding to P_n coloring configurations. Using U derived in the *Section* 2, we have:

$$P_n + P_{n-1} = k(k-1)^{n-1} \tag{26}$$

Which is a first-order linear recurrence relation with constant coefficients and a non-homogeneous term. Solving for this equation and we will get the same result.

4.2 Combination of Different Forms

In this case, we have to discuss in cases, which means we have to break the combined pattern into different pieces and discuss the different pieces perspectively and then use the counting principles. However, for most of the complex patterns, it's more easier to discuss from the view of Graph Theory, which can be introduced from transforming the patterns in to node-line graph (Figure 5). I personally don't know anything about this field, so I can't put too many comments here.